

Jagadeesh Rachapudi

M.S. Student @ IIT Mandi

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✍ About Me

I am a second-year M.S. student at **IIT Mandi** specializing in AI/ML research at the **Center for Artificial Intelligence**. My research focuses on Machine Unlearning, Large Language Models, and AI Safety under **Dr. Amit Shukla** and **Dr. Praful Hambarde**. I have strong Python development skills and experience building AI systems for model evaluation, security testing, and continual learning. I am passionate about developing reliable and safe AI systems.

🎓 Education

Master of Technology (Research), Robotics
Indian Institute of Technology (IIT Mandi)

Jan 2024 – Present
CGPA: 7.09/10.0

Bachelor of Technology, Computer Science
Rise Krishna Sai Gandhi Group of Institutions

Mar 2018 – Feb 2022
CGPA: 7.21/10.0

🔧 Technical Skills

Programming Languages: Python, C++, Java

ML/DL Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn

LLM & GenAI: Fine-tuning (LoRA, QLoRA), Prompt Engineering, Model Evaluation, RAG Pipelines

Tools & Libraries: Git, Docker, Weights & Biases, NumPy, Pandas, Matplotlib

Specialized: Machine Unlearning, Continual Learning, Model Security, Backdoor Detection, NLP

📖 Publications

[C1] BackFlush: Knowledge-Free Backdoor Detection and Elimination with Watermark Preservation in Large Language Models

Jagadeesh Rachapudi, Ritali Vatsi, Pranav Singh, Praful Hambarde, Amit Shukla

International Joint Conference on Neural Networks (IJCNN), 2026 (Accepted, CORE B)

[C2] DEG-Net: An Efficient Deep Neural Network for Railway Track Crack Segmentation

*Pranav Singh, Ritali Vatsi, **Rachapudi Jagadeesh**, Vinita Shukla, Vikas Sharma, Praful Hambarde, Amit Shukla*

ACM/SIGAPP Symposium on Applied Computing (SAC), 2026 (Accepted, CORE B)

[C6] Autonomous Pipeline Inspection Using Lightweight Vision-Based UAV Tracking

*Naisarg Panya, Surya Prakash S.K, Ayush Gupta, Ashok Kumar Sivarathri, **Jagadeesh Rachapudi**, Praful Hambarde, Amit Shukla*

IEEE Transactions on Artificial Intelligence (TAI) (Accepted, Q1)

[C7] Text-Centric Visual Question Answering Under Visual Degradation: A Comparative Study of Modular and End-to-End Approaches

*Ritali Vatsi, Himani Sharma, **Jagadeesh Rachapudi**, Vikas Sharma, Amit Shukla, Pawan Goyal*

IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS), 2025 (Accepted, Workshop)

- [C8] **Audio-Visual Talking-Head Deepfake Detection Using Modality-Specific Forensic Cues**
Ritali Vatsi, Rachapudi Jagadeesh, Shruti Singh Baghel, Amit Shukla, Pawan Goyal, Sahil Vatsi
IEEE Region 10 Conference (TENCON), 2026 (Accepted, CORE C)
- [C3] **RePAIR: Interactive Machine Unlearning through Prompt-Aware Model Repair [arXiv]**
Jagadeesh Rachapudi, Pranav Singh, Ritali Vatsi, Sachin Chaudhary, Praful Hambarde, Amit Shukla
AAAI Conference on Artificial Intelligence (AAAI), 2027 (Under Review, CORE A*)
- [C4] **Camouflage Unlearning: Fast and Surgical Machine Unlearning for Biometric Systems**
Jagadeesh Rachapudi, Sachin Chaudhary, Praful Hambarde, Amit Shukla
IEEE Signal Processing Letters (Under Review, Q1)
- [C5] **BID-LoRA: A Parameter-Efficient Framework for Continual Learning and Unlearning [arXiv]**
Jagadeesh Rachapudi, Ritali Vatsi, Praful Hambarde, Amit Shukla
Transactions on Machine Learning Research (TMLR) (Under Review, Q2)
- [C9] **SeDT: Sentence-Transformer Decision-Transformer Conditioning for Multi-Turn Conversation Reliability [arXiv]**
Ramakrishna Vamsi Setti, Rachapudi Jagadeesh, Sachin Chaudhary, Praful Hambarde, Amit Shukla
EMNLP ACL Rolling Review (ARR), May 2026 (Under Review, CORE A*)
- [C10] **PEC-Nav: Self-Calibrating Frontier Search for Embodied Question Answering**
Ritali Vatsi, Jagadeesh Rachapudi, Praful Hambarde, Pawan Goyal, Amit Shukla
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2027 (Under Review, Round 2, CORE A)

Work Experience

Simhatel Technologies Pvt. Ltd. **Aug 2026 – Present**
AI Engineer
Building AI-enabled robotic perception and inspection systems for industrial and infrastructure monitoring

Research Experience

Center for Artificial Intelligence @ IIT Mandi **Jan 2024 – Present**
M.Tech. Student (Advisors: Dr. Amit Shukla & Dr. Praful Hambarde)
Conducting research on Machine Unlearning with focus on LLMs and biometric systems
Published/submitted 10 papers (5 accepted) at top-tier AI conferences and journals

Academic Service

Conference Reviewer

- ACM/SIGAPP Symposium on Applied Computing (SAC) 2026
- International Joint Conference on Neural Networks (IJCNN) 2026

Achievements

CVPR 2025 NTIRE Challenge on RAW Image Restoration - Track 1: Super Resolution

- Developed RawSRMamba: Mamba-Based Raw Image Super-resolution